Arc Length and Improper Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arc length**

Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

$y' = x^{1/2}$, so $L = \int_0^3 \sqrt{1+x}\,dx = \left[\frac{2}{3}(1+x)^{3/2}\right]_0^3 = \frac{2}{3}(8-1) = \frac{14}{3}$.

2 **[Foundational]** • **Target: Arc length**

Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.

$L = \int_0^{\pi} \sqrt{1+\cos^2 x}\,dx$.

3 **[Foundational]** • **Target: Improper integrals**

Evaluate $\int_1^{\infty} \frac{1}{x^2}\,dx$, or state that it diverges.

$\lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b = \lim_{b\to\infty} \left(1 - \frac{1}{b}\right) = 1$: converges to 1.

4 **[Foundational]** • **Target: Improper integrals**

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$ (improper at the lower limit).

$\lim_{a\to 0^+} [2\sqrt{x}]_a^1 = 2 - 0 = 2$: converges to 2.

5 **[Foundational]** • **Target: More arc length practice**

Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.

$y' = 3$: $L = \int_0^4 \sqrt{1+9}\,dx = 4\sqrt{10}$. Distance check: endpoints $(0, 1)$ and $(4, 13)$: $\sqrt{4^2 + 12^2} = \sqrt{16+144} = \sqrt{160} = 4\sqrt{10}$. ✓

6 **[Foundational]** • **Target: The p-integral test**

For $\int_1^{\infty} \frac{1}{x^p}\,dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).

Converges when $p > 1$. As $b \to \infty$, $b^{1-p} \to 0$ only when $1 - p < 0$, i.e. $p > 1$; otherwise the limit is infinite (or, for $p = 1$, the antiderivative is $\ln x$, which also diverges).

7 **[Foundational]** • **Target: The p-integral test**

Use the p-integral result to quickly determine whether $\int_1^{\infty} \frac{1}{x^{0.9}}\,dx$ converges.

$p = 0.9 < 1$: diverges.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Improper integrals with infinite discontinuities inside the interval**

Evaluate $\int_0^2 \frac{1}{(x-1)^2}\,dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).

Split at $x = 1$: $\int_0^1 \frac{1}{(x-1)^2}dx + \int_1^2 \frac{1}{(x-1)^2}dx$. Near $x = 1$ from either side, $\lim_{b\to 1}\left[-\frac{1}{x-1}\right]$ blows up to $\pm\infty$: both pieces diverge, so the full integral diverges.

9 **[Intermediate]** • **Target: Improper integrals with infinite discontinuities inside the interval**

Evaluate $\int_{-1}^1 \frac{1}{x^{1/3}}\,dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.

By symmetry (odd integrand), $\int_{-1}^0 x^{-1/3}dx = -\int_0^1 x^{-1/3}dx$ (each piece converges since $p = \frac{1}{3} < 1$ for this type of integrand near a finite discontinuity). The two pieces cancel: the integral equals 0.

10 **[Intermediate]** • **Target: Visualizing an improper integral as a limiting area**

The graph shows $f(x) = \frac{1}{x^2}$ on $[1, 10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (found earlier).

The curve $y = \frac{1}{x^2}$ decreases rapidly toward 0, so the strips added as x grows contribute less and less area. Even though the region extends infinitely far right, the total shaded area converges to a finite value, 1, because the tail contributions shrink fast enough (consistent with the p-test, $p = 2 > 1$).

11 **[Intermediate]** • **Target: Arc length of a more involved curve**

Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.

$y' = 2x$: $L = \int_0^2 \sqrt{1 + 4x^2}\,dx$. Straight-line distance between $(0, 0)$ and $(2, 4)$: $\sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5} \approx 4.47$, which is a lower bound since the actual curve is longer than the chord connecting its endpoints.

12 **[Intermediate]** • **Target: Comparison test for improper integrals**

Use a comparison with $\int_1^\infty \frac{1}{x^2}\,dx$ (known to converge) to argue that $\int_1^\infty \frac{1}{x^2+1}\,dx$ also converges, without evaluating it.

For $x \geq 1$, $\frac{1}{x^2+1} < \frac{1}{x^2}$. Since $\int_1^\infty \frac{1}{x^2}\,dx$ converges (to 1), and the integrand of interest is smaller and positive everywhere on $[1, \infty)$, the comparison test guarantees $\int_1^\infty \frac{1}{x^2+1}\,dx$ also converges.

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13 **[Intermediate]** • **Target: Comparison test for improper integrals**

Use a comparison with $\int_1^\infty \frac{1}{x}\,dx$ (known to diverge) to argue that $\int_1^\infty \frac{1}{\sqrt{x}}\,dx$ also diverges.

For $x \geq 1$, $\frac{1}{\sqrt{x}} \geq \frac{1}{x}$ (since $\sqrt{x} \leq x$). Since $\int_1^\infty \frac{1}{x}\,dx$ diverges and the integrand of interest is even larger, the comparison test guarantees $\int_1^\infty \frac{1}{\sqrt{x}}\,dx$ also diverges.

14 **[Intermediate]** • **Target: A capstone arc-length application**

A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).

$y' = \frac{x}{4}$: $L = \int_{-4}^{4} \sqrt{1 + \frac{x^2}{16}}\,dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A capstone arc-length application** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x-axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.

$y' = \frac{-x}{\sqrt{25 - x^2}}$: $L = \int_{-4}^{4} \sqrt{1 + \frac{x^2}{25 - x^2}}\,dx = \int_{-4}^{4} \sqrt{\frac{25}{25 - x^2}}\,dx = \int_{-4}^{4} \frac{5}{\sqrt{25 - x^2}}\,dx$.

16 **[Challenging]** • **Target: Improper integrals with infinite discontinuities inside the interval** • **Cross-Lesson: 4.02 Riemann Sums and Definite Integrals**

Level 2 · MediumApply the Ratio Test to determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{4^n}{n!}$.

Set up the ratio: $\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{4^{n+1}}{(n+1)!} \cdot \frac{n!}{4^n}\right| = \lim_{n\to\infty} \left(\frac{4^{n+1}}{4^n} \cdot \frac{n!}{(n+1)n!}\right) = \lim_{n\to\infty} \frac{4}{n+1} = 0.$ Since \(\rho = 0 Ratio Test, the series converges absolutely.

17 **[Challenging]** • **Target: Visualizing an improper integral as a limiting area** • **Cross-Lesson: 5.01 The Fundamental Theorem of Calculus**

[Unit 9 · Polar Coordinates & Polar Area · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 **[Challenging]** • **Target: Arc length of a more involved curve** • **Cross-Lesson: 6.01 Related Rates**

[Unit 10 · Infinite Sequences & Series — Convergence Tests · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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19 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 6.02 Optimization and Curve Sketching

Level 1 · FoundationsFind all values of c in the open interval $(-1, 2)$ that satisfy the Mean Value Theorem for $f(x) = x^3 - 2x^2 + 1$.

1. Check hypotheses: $f(x)$ is a polynomial, continuous on $[-1, 2]$ and differentiable on $(-1, 2)$. 2. Calculate the average rate of change on $[-1, 2]$: $f(-1) = (-1)^3 - 2(-1)^2 + 1 = -1 - 2 + 1 = -2$ $f(2) = 2^3 - 2(2)^2 + 1 = 8 - 8 + 1 = 1$ $\frac{f(2) - f(-1)}{2 - (-1)} = \frac{1 - (-2)}{3} = \frac{3}{3} = 1$ 3. Set $f'(c) = 1$: $f'(x) = 3x^2 - 4x \Rightarrow 3c^2 - 4c = 1 \Rightarrow 3c^2 - 4c - 1 = 0$ Apply the quadratic formula.

20 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 7.01 Integration by Substitution

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

(a) Under H_0, $X \sim \text{Po}(6)$. Calculate cumulative probabilities: $P(X = 0) = e^{-6} \approx 0.00248$ $P(X \le 1) = e^{-6}(1 + 6) = 7e^{-6} \approx 0.01735$ $P(X \le 2) = e^{-6}(1 + 6 + 18) = 25e^{-6} \approx 0.06197$ Since $P(X \le 1) 0.05$, the critical region is $X \le 1$ (i.e. $X = 0$ or 1). (b) Actual significance level = $P(X \le 1 \mid \lambda = 6) = 7e^{-6} \approx 0.0174$ (or 1.74%). (c) Type II error occurs when we fail to reject H_0 when $\lambda = 3$. We fail to reject when.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arc length

Comparison test for improper integrals: $0 \le f(x) \le g(x)$, $\int_a^\infty g$ converges implies:

a. $\int_a^\infty f$ **converges** ✓

b. $\int_a^\infty f$ diverges

c. $f = g$

d. nothing

Comparison: dominated convergent integral converges.

22 [Foundational] • Target: Improper integrals

If $f'(x) = (x - 3)$, on which interval is f increasing?

a. $(3, \infty)$ ✓

b. \{3\}

c. $(-\infty, \infty)$

d. $(-\infty, 3)$

Applying Application of Derivatives principles, $f'(x) > 0$ when $x - 3 > 0$, i.e. $x > 3$. Therefore, the verified correct option is $(3, \infty)$.

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23 [Foundational] • Target: More arc length practice

Polar area inside $r = 1 + \cos\theta$ and outside $r = 1$:

a. $2 + \pi/4$ ✓

b. $\pi/4 + 1$

c. $1 + \pi/4$

d. $\pi/2$

$A = \frac{1}{2}\int_{-\pi/2}^{\pi/2}[(1+\cos\theta)^2 - 1]d\theta = \frac{1}{2}\int(2\cos\theta + \cos^2\theta)d\theta = 2 + \pi/4.$

24 [Foundational] • Target: More improper integrals

Polar: find the area inside $r = 3\sin\theta$ and outside $r = 1 + \sin\theta$:

a. $\pi/4$

b. π ✓

c. $\pi/2$

d. $3\pi/4$

Intersect: $3\sin\theta = 1 + \sin\theta \Rightarrow \sin\theta = 1/2 \Rightarrow \theta = \pi/6, 5\pi/6$; $A = \pi$.

25 [Foundational] • Target: The p-integral test

A box with square base and open top must hold 32 m³. Minimise surface area. Optimal side length x:

a. 2

b. 4 ✓

c. $2\sqrt[3]{4}$

d. $\sqrt[3]{32}$

$S = x^2 + 4x(32/x^2) = x^2 + 128/x$; $S' = 2x - 128/x^2 = 0 \Rightarrow x^3 = 64 \Rightarrow x = 4$.

26 [Foundational] • Target: Improper integrals with infinite discontinuities inside the interval

Area inside $r = 2\cos\theta$ and outside $r = 1$:

a. $\frac{\pi}{2} + \frac{\sqrt{3}}{2}$

b. $\frac{\pi}{3} + \frac{\sqrt{3}}{2}$ ✓

c. $\pi/3$

d. $\pi/2$

Intersection at $\cos\theta = 1/2 \Rightarrow \theta = \pm\pi/3$; $A = \pi/3 + \sqrt{3}/2$.

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27 [Foundational] • Target: Visualizing an improper integral as a limiting area

If $f'(x) > 0$ on an interval, f is:

a. concave down

b. decreasing

c. **increasing ✓**

d. constant

A positive derivative means the function is increasing on that interval.

28 [Foundational] • Target: Arc length of a more involved curve

If $f''(x) > 0$ on an interval, the graph of f is:

a. **concave up ✓**

b. decreasing

c. concave down

d. linear

A positive second derivative means the graph curves upward (concave up).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Comparison test for improper integrals

Arc length of $r = 2\cos\theta$ (full circle, $0 \le \theta \le \pi$):

a. π

b. 4

c. $\pi/2$

d. 2π ✓

Circle radius 1 (centered at $(1, 0)$), circumference $= 2\pi$.

30 [Intermediate] • Target: A capstone arc-length application

For the catenary $y = a\cosh(x/a)$, the arc length from $-L$ to L is:

a. $2L$

b. $2a\sinh(L/a)$ ✓

c. $L\cosh(L/a)$

d. $2a\cosh(L/a)$

$y' = \sinh(x/a)$; $L = \int_{-L}^{L} \cosh(x/a)dx = 2a\sinh(L/a)$.

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31 [Intermediate] • Target: Arc length

Area swept inside one loop of $r^2 = 4\sin 2\theta$:

a. 1

b. 2 ✓

c. 4

d. π

$A = \frac{1}{2}\int_0^{\pi/2} 4\sin 2\theta \, d\theta = 2$.

32 [Intermediate] • Target: Improper integrals

Comparison test: if $0 \le a_n \le b_n$ and $\sum b_n$ converges, then $\sum a_n$:

a. diverges

b. oscillates

c. **converges** ✓

d. is zero

Direct comparison: dominated by convergent.

33 [Intermediate] • Target: More arc length practice

$\int_0^{\pi}\int_0^{\sin\theta} r\,dr\,d\theta$: inner integral gives $\sin^2\theta/2$; outer gives

a. π

b. $\pi/2$

c. $\pi/4$ ✓

d. $1/2$

$\int_0^{\pi} \sin^2\theta/2 \, d\theta = \pi/4$.

34 [Intermediate] • Target: More improper integrals

Triple integral in cylindrical: $\int_0^{2\pi}\int_0^1\int_0^1 r\,dz\,dr\,d\theta =$

a. 4π

b. 2π

c. $\pi/2$

d. π ✓

$= 2\pi\int_0^1 r\,dr = 2\pi \cdot 1/2 = \pi$.

35 [Intermediate] • Target: The p-integral test

Arc length of helix $\mathbf{r}(t) = (\cos t, \sin t, t)$, $0 \le t \le 2\pi$:

a. 2π

b. $\sqrt{2}$

c. 4π

d. $2\pi\sqrt{2}$ ✓

$L = \sqrt{2} \cdot 2\pi = 2\pi\sqrt{2}$.

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36 [Intermediate] • Target: Improper integrals with infinite discontinuities inside the interval

Arc length parametric: $L = \int_a^b \sqrt{\dot{x}^2 + \dot{y}^2}\, dt$. For $x = t, y = t$ from 0 to 1:

a. $\sqrt{2}$ ✓

b. 1

c. 2

d. $\sqrt{2}/2$

$L = \int_0^1 \sqrt{1+1}\, dt = \sqrt{2}$.

37 [Intermediate] • Target: Visualizing an improper integral as a limiting area

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$ ✓

d. $\frac{2}{3}$

Integrating over the triangle $x \ge 0, y \ge 0, x + y \le 1$ gives $\int_0^1 \int_0^{1-x} 6xy\, dy\, dx = 3\int_0^1 x(1-x)^2\, dx = 3\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{1}{4}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Arc length of a more involved curve • Cross-Lesson: 6.01 Related Rates

Use $x = u + v, y = u - v$ to compute $\iint_D (x - y)e^{x+y}\, dA$ where D is the square $|x - y| \le 1, |x + y| \le 1$:

a. 0 ✓

b. $2(e - e^{-1})$

c. $(e - 1)^2$

d. $4(e - e^{-1})/e$

Under the change of variables $2u = x + y$ and $2v = x - y$, the integrand $2v\, e^{2u}$ is odd in v over the symmetric interval $[-1/2, 1/2]$, so the double integral equals 0.

39 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 6.02 Optimization and Curve Sketching

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$ ✓

d. $-\frac{3}{2}$

$x^2 + y^2 = 10^2$. At $x = 6$: $y = \sqrt{10^2 - 6^2} = 8$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{3}{4}$ m/s.

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40 [Challenging] • Target: A capstone arc-length application • Cross-Lesson: 7.01 Integration by Substitution

Evaluate $\iiint_B (x^2+y^2+z^2)\,dV$ where B is the ball $x^2+y^2+z^2 \le 1$.

a. $\frac{4\pi}{5}$ ✓

b. $\frac{4\pi}{3}$

c. $\frac{2\pi}{5}$

d. $\frac{8\pi}{5}$

In spherical coordinates the integrand is ρ^2, so $\int_0^{2\pi}\int_0^{\pi}\int_0^{1}\rho^4\sin\phi\,d\rho\,d\phi\,d\theta = \frac{4\pi}{5}$.

41 [Challenging] • Target: Arc length • Cross-Lesson: 7.02 Integration by Parts

Variation of parameters for $y''+y=\sec x$: general solution involves $\ln|\sec x+\tan x|$. This appears because the particular integral of $\int \sin x \sec x\,dx =$

a. $\tan x$

b. $\ln|\cos x|$

c. $-\ln|\cos x|$ ✓

d. $-\cos x$

Applying Differential Equations principles, $\int \tan x\,dx = -\ln|\cos x|$. Therefore, the verified correct option is $-\ln|\cos x|$.

42 [Challenging] • Target: Improper integrals • Cross-Lesson: 8.01 Area Between Curves

The Jacobian of the transformation $x = r\cosh u,\ y = r\sinh u$ is r times:

a. $\cosh^2 u - \sinh^2 u = 1$ ✓

b. $\cosh u + \sinh u$

c. $\sinh^2 u$

d. r^2

Applying Multiple Integrals principles, $J = r(\cosh^2 u - \sinh^2 u) = r$. Therefore, the verified correct option is $\cosh^2 u - \sinh^2 u = 1$.

43 [Challenging] • Target: More arc length practice • Cross-Lesson: 8.02 Volumes of Revolution

Surface integral $\iint_S c\,dS$ for constant c equals:

a. $c \cdot \text{Area}(S)$ ✓

b. c

c. 0

d. $c \cdot \text{Vol}$

Applying Vector Calculus principles, $\iint_S c\,dS = c \cdot A(S)$. Therefore, the verified correct option is $c \cdot \text{Area}(S)$.

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44 [Challenging] • Target: More improper integrals • Cross-Lesson: 9.01 Separable Differential Equations

Centre of mass of lamina $D=[0,1]\times[0,1]$ with density $\delta=xy$:

a. $(1/2,1/2)$

b. $(2/3,2/3)$ ✓

c. $(3/4,3/4)$

d. $(1,1)$

$m=\int_0^1\int_0^1 xy\,dy\,dx=1/4$; $M_y=\int_0^1\int_0^1 x^2y\,dy\,dx=1/6$; $\bar{x}=(1/6)/(1/4)=2/3$; similarly $\bar{y}=2/3$.

45 [Challenging] • Target: The p-integral test • Cross-Lesson: 1.01 Precalculus Review for Calculus

Use polar coordinates to evaluate $\iint_R (x^2+y^2)^3\,dA$, where R is the disk $x^2+y^2\le 9$.

a. $\frac{6561}{4}$

b. $\frac{6561\pi}{4}$ ✓

c. $\frac{6561\pi}{2}$

d. $\frac{6561\pi}{8}$

With $x=r\cos\theta,\ y=r\sin\theta$ and $dA=r\,dr\,d\theta$, the integral becomes elementary; its value is $\frac{6561\pi}{4}$.

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